QML Lab • research project 03

Projected vs Fidelity Kernels

How fast do quantum kernels concentrate as qubits are added, and can projected kernels delay it?

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Quadrant	CQ (classical data, quantum processor)	**Track**	Kernels
Difficulty	Hard	**Tools**	PennyLane, NumPy
Builds on	Modules 6 and 8 of the QML Lab	**Workspace**	`qml_projects/projected-vs-fidelity-kernels/`

Summary. Quantum kernels are attractive because they use exponentially large feature spaces, but that same size makes kernel values concentrate: all pairs of points start to look equally (dis)similar. This project measures concentration for fidelity kernels and projected kernels from 2 to 12 qubits, for different feature maps and bandwidths, and with finite measurement shots. The result is a map of where quantum kernels remain usable.

1 Background and key concepts

1.1 Fidelity and projected kernels

The fidelity kernel compares whole states, $\kappa^F(x,x') = \mathrm{Tr}[\rho(x)\rho(x')]$. The projected kernel of Huang et al. (2021) compares only reduced single-qubit states $\rho_k(x)$:

$$\kappa^{PQ}(x,x') = \exp\Big(-\gamma \sum_{k=1}^{n} \|\rho_k(x) - \rho_k(x')\|_F^2\Big), \qquad \|\rho_k - \rho_k'\|_F^2 = \tfrac{1}{2} \sum_{P\in\{X,Y,Z\}} \big(\langle P_k\rangle - \langle P_k\rangle'\big)^2. \quad (1)$$

It needs only single-qubit expectation values, which are cheap to measure, and it behaves like an RBF kernel on a learned, quantum-generated feature vector.

1.2 Exponential concentration

Thanasilp et al. (2024) prove that for expressive embeddings, entangling embeddings, global measurements or noise, the kernel values concentrate around a fixed value with variance

$$\mathrm{Var}_{x,x'}\big[\kappa(x,x')\big] \in O(b^{-n}), \qquad b > 1. \quad (2)$$

With M shots, a kernel entry is estimated with standard error about $\sqrt{\kappa(1-\kappa)/M}$, so once the spread of kernel values falls below this error the Gram matrix is dominated by shot noise. Projected kernels avoid the global overlap but also concentrate when reduced states approach the maximally mixed state.

1.3 Bandwidth and inductive bias

Scaling features by a bandwidth λ keeps encoded states closer together and delays concentration (Shaydulin and Wild, 2022). Kübler et al. (2021) argue more generally that useful quantum

kernels must restrict themselves to a small, task-relevant part of Hilbert space. A related warning is that settings that provably avoid concentration are often the ones that classical computers can simulate (Cerezo et al., 2023).

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 At which qubit count does the off-diagonal variance of each kernel drop below the shot-noise level for $M = 10^3$ and 10^4 shots?

RQ2 Does reducing the bandwidth delay concentration equally for fidelity and projected kernels?

RQ3 How do the results depend on the feature map (IQP/ZZ versus a deep hardware-efficient embedding)?

3 Method

3.1 Data

Synthetic inputs drawn uniformly from $[0, 2\pi]^n$ to isolate the effect of the embedding, and a real data set (breast cancer, PCA to n features) to see the effect on classification. For each qubit number $n = 2, \ldots, 12$, 200 random pairs estimate the kernel statistics.

3.2 Models

Exact state-vector simulation of the IQP/ZZ map and of a hardware-efficient embedding with $L = n$ layers. Fidelity kernels from state overlaps; projected kernels from single-qubit Pauli expectations with γ set by the median heuristic. Finite-shot versions sample from the exact probabilities. For classification, both kernels feed a precomputed-kernel SVM.

3.3 Baselines

An RBF kernel on the same inputs (with tuned width) for the classification part; the analytic $O(2^{-n})$ reference curve for Haar-random states.

3.4 Metrics

- Mean and variance of off-diagonal kernel values versus n (log scale).
- Effective rank of the Gram matrix.
- Test accuracy of the SVM and geometric difference to the RBF kernel.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Concentration curves	$n = 2..12$, both kernels, IQP map	off-diagonal variance
E2	Bandwidth	λ sweep at each n	variance, accuracy
E3	Feature map	IQP vs hardware-efficient	variance slope
E4	Finite shots	$M \in \{10^2, 10^3, 10^4\}$	rank, accuracy
E5	Classification	breast cancer, $n = 2..10$	accuracy, g

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/projected-vs-fidelity-kernels/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/projected-vs-fidelity-kernels/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "concentration"
title = "Kernel concentration from 2 to 12 qubits"
script = "experiments/e01_concentration.py"
figure = "concentration.png"

[experiments.params]
max_qubits = 12
n_pairs = 200
bandwidth = 1.0
feature_map = "iqp"
shots = 0
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/projected-vs-fidelity-kernels
# run by hand with the default parameters
```

```
PYTHONPATH=../../qml_lab python experiments/e01_concentration.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Plot variances on a log scale and fit the slope; report the fitted base b with its uncertainty.

7 Open directions

- k-body projected kernels that use reduced states of $k > 1$ qubits.
- Trainable bandwidth (connects to Project 02).
- Check whether the non-concentrating regimes can be simulated classically (Cerezo et al., 2023).

8 Risks and limitations

- Conclusions depend on the input distribution; uniform synthetic data is a worst case.
- The "shot-noise threshold" depends on the chosen M; report results for several values.
- State-vector memory limits the study to about 14 qubits on a laptop.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

Cerezo, M., Larocca, M., García-Martín, D., Diaz, N. L., Braccia, P. et al. (2023) 'Does provable absence of barren plateaus imply classical simulability? Or, why we need to rethink variational quantum computing', Preprint, arXiv:2312.09121. Available at: https://arxiv.org/abs/2312.09121.

Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H. and McClean, J. R. (2021) 'Power of data in quantum machine learning', *Nature Communications*, 12, 2631. Available at: https://doi.org/10.1038/s41467-021-22539-9.

Kübler, J. M., Buchholz, S. and Schölkopf, B. (2021) 'The inductive bias of quantum kernels', in *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp. 12661–12673.

Shaydulin, R. and Wild, S. M. (2022) 'Importance of kernel bandwidth in quantum machine learning', *Physical Review A*, 106(4), 042407. Available at: https://doi.org/10.1103/PhysRevA.106.042407.

Thanasilp, S., Wang, S., Cerezo, M. and Holmes, Z. (2024) 'Exponential concentration in quantum kernel methods', *Nature Communications*, 15, 5200. Available at: https://doi.org/10.1038/s41467-024-49287-w.